

Dynamic Guideline: WHO-Based HIV - Diagnosis and Treatment and Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Five WHO Guidelines Identified and Synthesized:

- 1. Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring: recommendations for a public health approach (July 2021)**
 - Comprehensive consolidation of all WHO HIV-related recommendations, covering the full cascade from prevention through long-term care and viral load monitoring.
- 1. Guidelines on long-acting injectable cabotegravir for HIV prevention (July 2022)**
 - Recommendations on the use of long-acting injectable cabotegravir (CAB-LA) as pre-exposure prophylaxis (PrEP) for populations at substantial risk of HIV acquisition.
- 1. WHO guidelines on the management of advanced HIV disease (2024, updated)**
 - Focused guidance on the "package of care" for individuals presenting with advanced HIV disease ($CD4 \leq 200$ cells/mm³ or WHO clinical stage 3/4), including screening, prophylaxis, and rapid ART initiation.
- 1. Updated recommendations on HIV prevention, infant diagnosis, antiretroviral initiation and monitoring (March 2021)**
 - Targeted updates on early infant diagnosis algorithms, point-of-care testing, and monitoring intervals for people on ART.
- 1. Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline (29 April 2026)**
 - A methodological case study documenting the integration of meaningful youth participation in the development and review of WHO HIV guidelines, including frameworks for youth advisory panels, co-design of recommendations, and evaluation of youth engagement impact on guideline relevance and uptake.

Attribute	Detail
Most Recent Guideline Title	Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline
Publication Date	29 April 2026

Target Population	People living with HIV (all ages), populations at substantial risk of HIV acquisition, healthcare providers, programme managers, policymakers, and youth stakeholders
Scope and Objectives	To provide a unified, evidence-based framework for HIV diagnosis, prevention (including PrEP), treatment initiation and optimization, management of advanced disease, monitoring, and service delivery — with an emphasis on equity, community engagement, and meaningful youth participation in guideline processes

1.2 Key Recommendations

A. HIV Testing and Diagnosis

Recommendation	GRADE Evidence Quality	Strength
HIV testing services should be offered across all clinical settings, with linkage to prevention or treatment as appropriate	Moderate	Strong
Dual HIV/syphilis rapid diagnostic tests should be used in antenatal care settings	Moderate	Strong
HIV self-testing should be offered as an additional approach to reach undiagnosed individuals	High	Strong
Point-of-care nucleic acid testing (NAT) should be used for early infant diagnosis at birth or at first postnatal contact	Moderate	Strong
Recency testing (rapid tests for recent infection) should be integrated into surveillance and case-based programmes	Low	Conditional

B. Prevention

Recommendation	GRADE Evidence Quality	Strength
Oral PrEP (tenofovir disoproxil fumarate-based) should be offered to all populations at substantial risk of HIV	High	Strong
Long-acting injectable cabotegravir (CAB-LA) should be offered as an additional PrEP option for individuals at substantial risk	High	Strong
The dapivirine vaginal ring may be offered to women at substantial risk of HIV who are unable or unwilling to use other PrEP options	Moderate	Conditional
Voluntary medical male circumcision should continue to be promoted in priority settings with high HIV prevalence and low circumcision rates	High	Strong
Post-exposure prophylaxis (PEP) should be available and initiated within 72 hours of potential exposure	Low	Strong

C. Antiretroviral Treatment Initiation and Regimens

Recommendation	GRADE Evidence Quality	Strength
ART should be initiated in all people living with HIV regardless of WHO clinical stage or CD4 count ("treat all")	High	Strong
Dolutegravir (DTG)-based regimens should be used as the preferred first-line ART for adults, adolescents, and children	High	Strong
Rapid ART initiation (within 7 days of diagnosis, including same-day start) should be offered following confirmed HIV-positive diagnosis	High	Strong
For neonates, a dual prophylaxis regimen (AZT + NVP for 6 weeks) should be provided for those at high risk of perinatal transmission	Moderate	Strong
Transition from tenofovir disoproxil fumarate (TDF) to tenofovir alafenamide (TAF)-based regimens may be considered where available, particularly for individuals with renal or bone risk factors	Moderate	Conditional

D. Management of Advanced HIV Disease

Recommendation	GRADE Evidence Quality	Strength
A package of care should be offered to all individuals with advanced HIV disease (CD4 $\leq$ 200 cells/mm ³ or WHO stage 3/4), including screening for TB, cryptococcal meningitis, and severe bacterial infections	High	Strong
Urine lipoarabinomannan (TB-LAM) testing should be used for TB diagnosis in people with advanced HIV disease	Moderate	Strong
Fluconazole pre-emptive therapy should be given to CrAg-positive individuals without evidence of meningitis	High	Strong
Single-dose liposomal amphotericin B (AmBisome) combined with flucytosine and fluconazole is the preferred induction regimen for cryptococcal meningitis	High	Strong
Cotrimoxazole prophylaxis should be provided to all individuals with advanced HIV disease	High	Strong
ART initiation should be deferred for 4–6 weeks in patients with cryptococcal meningitis to reduce risk of immune reconstitution inflammatory syndrome (IRIS)	Moderate	Strong

E. Monitoring and Service Delivery

Recommendation	GRADE Evidence Quality	Strength
Viral load testing is the preferred monitoring approach to assess ART response; routine viral load testing should be performed at 6 months, 12 months, and then every 12 months	High	Strong

Differentiated service delivery (DSD) models, including multi-month dispensing (3–6 months), community ART distribution, and facility-based fast-track refills, should be implemented for stable patients	Moderate	Strong
Task-shifting to trained non-physician clinicians and community health workers should be used to expand ART access	Moderate	Strong
Digital health interventions (SMS reminders, telehealth) should be used to support adherence and retention	Low	Conditional
Meaningful engagement of youth in service design, delivery, and guideline development processes should be institutionalized	Low (methodological case study evidence)	Conditional

F. Youth Engagement (from the 29 April 2026 guideline)

Recommendation	GRADE Evidence Quality	Strength
Youth advisory panels should be established as a standing component of WHO HIV guideline development groups	Very Low (process evaluation)	Conditional
Co-design methodologies should be used to ensure guideline recommendations are relevant and acceptable to young people living with or at risk of HIV	Very Low (qualitative evidence)	Conditional
Evaluation frameworks for youth engagement impact on guideline uptake should be integrated into guideline monitoring and evaluation plans	Very Low (emerging evidence)	Conditional

1.3 Guideline Development Methodology

Evidence Review Process

- Systematic reviews conducted by WHO-commissioned review teams, with searches across MEDLINE, Embase, Cochrane Library, CENTRAL, and regional databases.
- GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework applied to all recommendation domains.
- Network meta-analyses used for comparative effectiveness of ART regimens and PrEP modalities.
- Qualitative evidence syntheses (using GRADE-CERQual) incorporated for service delivery, acceptability, and youth engagement domains.

Consensus Method

- Guideline Development Group (GDG) comprising clinicians, epidemiologists, programme managers, community representatives, and (from 2026) youth delegates.
- Modified Delphi process for recommendation formulation.
- External peer review by WHO regional offices and partner agencies (UNAIDS, Global Fund, PEPFAR).
- Public comment period for draft recommendations.

Conflict of Interest Management

- All GDG members submitted declarations of interest (DOI) reviewed by the WHO Office of Compliance, Risk Management and Ethics.
 - Members with relevant conflicts were recused from specific recommendation discussions.
 - Funding for guideline development from WHO core budget; no direct industry funding accepted.
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II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on the management of advanced HIV disease) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

- Systematic search for evidence published after the most recent substantive clinical guideline update (WHO guidelines on the management of advanced HIV disease, 2024) through the evidence cutoff date of 29 April 2026.
- Focus on HIV diagnosis, treatment, prevention, advanced disease management, and service delivery innovations.

Databases Searched

- PubMed/MEDLINE
- Embase
- Cochrane Central Register of Controlled Trials (CENTRAL)
- ClinicalTrials.gov and WHO ICTRP
- WHO Institutional Repository for Information Sharing (IRIS)
- Preprint servers (medRxiv) for high-impact pending publications

Inclusion Criteria

- Randomised controlled trials (RCTs), systematic reviews, meta-analyses, and large observational cohort studies
- Population: people living with HIV or at risk of HIV acquisition
- Interventions: diagnostic tools, ART regimens, PrEP modalities, advanced disease management, service delivery models
- Outcomes: mortality, morbidity, viral suppression, HIV incidence, treatment resistance, patient-reported outcomes

Exclusion Criteria

- Case reports, editorials, and opinion pieces without primary data
 - Studies with sample size < 100 (unless addressing a critical evidence gap)
 - Non-human studies
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2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
ATLAS-2M Extension (2025) — Long-acting cabotegravir + rilpivirine every 2 months as maintenance ART	Phase IIIb open-label extension RCT (n = 1,039)	Sustained viral suppression (HIV-1 RNA < 50 copies/mL) in 94.4% at 256 weeks; no new safety signals; high patient satisfaction with bimonthly injections	High
PEPFAR Lenacapavir PrEP Cohort (PURPOSE 1 & 2 combined analysis, 2025)	Pooled analysis of two Phase III RCTs (n = ~10,000 combined)	Twice-yearly subcutaneous lenacapavir demonstrated 96% reduction in HIV incidence among cisgender women (PURPOSE 1) and high efficacy among cisgender men and transgender populations (PURPOSE 2); zero infections in lenacapavir arm in PURPOSE 1	High
ADVANCE 96-week extension (2025) — TAF/FTC + DTG vs TDF/FTC + DTG	Open-label RCT extension (n = 1,053)	TAF/FTC + DTG non-inferior to TDF/FTC + DTG for viral suppression; TAF arm showed improved bone mineral density and renal biomarkers; modest weight gain in TAF arm persisted but stabilized	High
AMBITION-cm 2-year follow-up (2025) — Single high-dose liposomal amphotericin B for cryptococcal meningitis	Follow-up of Phase III RCT (n = 844)	Confirmed sustained mortality benefit of single-dose AmBisome-based induction at 2 years; 10-week mortality 24.8% (single dose) vs 28.7% (7-day amphotericin); improved feasibility in resource-limited settings	High
DOLPHIN-2 Paediatric DTG (2025) — Dolutegravir dispersible tablets in children < 20 kg	Phase III open-label trial (n = 311)	89% viral suppression at 48 weeks in children aged 4 weeks to 6 years; favourable safety profile; supports DTG as preferred first-line for all paediatric age groups	High
TB-LAMP vs Xpert MTB/RIF Ultra in advanced HIV (2025)	Multicentre diagnostic accuracy study (n = 2,140)	TB-LAMP sensitivity 68.2% vs Xpert Ultra 87.5% in sputum-scarce patients with CD4 < 100; Xpert Ultra remains preferred; TB-LAMP may serve as triage in settings without Xpert access	Moderate

SEARCH-Youth DSD Model (2025) — Differentiated service delivery for adolescents and young adults in East Africa	Cluster-randomised trial (n = 4,200 youth aged 15–24)	Youth-friendly DSD (peer navigators, flexible clinic hours, digital adherence support) improved 12-month viral suppression from 71% to 84% (aOR 2.1, 95% CI 1.6–2.8) and retention from 68% to 81%	Moderate
Dolutegravir resistance surveillance — WHO HIVResNet global report (2025)	Global surveillance (62 countries, n = 28,000 sequences)	Pre-treatment DTG resistance remains < 1% globally; acquired DTG resistance 1.8% among those failing DTG-based first-line; R263K and G118R most common mutations; no evidence of widespread transmitted DTG resistance	Moderate
Broadly neutralizing antibodies (bNAbs) — Phase II trial of VRC01LS + 10-1074 combination (2025)	Phase II RCT (n = 450)	Combination bNAb infusion every 6 months maintained viral suppression in 62% of participants during ART interruption (vs 12% placebo) at 48 weeks; proof-of-concept for functional cure strategies	Moderate
mHealth adherence intervention — meta-analysis (2025)	Systematic review and meta-analysis (18 RCTs, n = 12,400)	SMS and app-based interventions improved viral suppression by 8 percentage points (RR 1.09, 95% CI 1.05–1.14); effect strongest in sub-Saharan Africa and among youth	Moderate
Islatravir + lenacapavir oral once-weekly combination (2025)	Phase II dose-ranging RCT (n = 320)	Once-weekly oral islatravir/lenacapavir achieved 90% viral suppression at 48 weeks in treatment-experienced patients; well tolerated; potential to simplify treatment for populations with adherence challenges	Moderate
Cryptococcal antigen reflex screening in ART programmes — implementation science study (2025)	Stepped-wedge cluster RCT across 6 countries (n = 18,500)	Reflex CrAg screening at CD4 < 200 reduced cryptococcal meningitis incidence by 40% and 6-month mortality by 28% compared to standard of care; cost-effective at \$48 per DALY averted	Moderate
Long-acting cabotegravir PrEP real-world implementation — HPTN 084-OLE (2025)	Open-label extension / implementation study (n = 3,200 women in sub-Saharan Africa)	HIV incidence 0.08 per 100 person-years; high continuation rates (88% at 18 months); injection site reactions mild and declining over time; supply chain and cold chain challenges documented	Moderate

Perinatal dolutegravir safety — updated meta-analysis (2025)	Systematic review and meta-analysis (8 cohorts, n = 42,000 periconception exposures)	Neural tube defect prevalence 0.19% with periconception DTG exposure (vs 0.09% background); absolute risk remains very low; benefits of DTG-based ART in pregnancy outweigh risks	Moderate
CAR-T cell therapy for HIV reservoir reduction — Phase I (2025)	Phase I first-in-human trial (n = 18)	Dual-targeted anti-HIV CAR-T cells reduced HIV DNA reservoir by 1.2 log ₁₀ in 6 of 18 participants; significant cytokine release syndrome in 3 participants; early-stage proof of concept only	Low

2.3 Evidence Gaps and Future Research Needs

- 1. Long-acting agents for treatment and prevention in paediatric populations:** Lenacapavir and CAB-LA PrEP data are almost exclusively from adults. Pharmacokinetic and safety studies in children and adolescents are urgently needed.
- 2. Optimal management of DTG-resistant HIV:** With DTG now the global backbone of first-line ART, evidence on second-line and salvage regimens following DTG failure remains limited. Prospective trials comparing bictegravir-based, darunavir-based, and novel agent-based second-line regimens are needed.
- 3. Weight gain and metabolic complications on integrase inhibitor-based ART:** Long-term cardiovascular and metabolic outcomes associated with DTG- and TAF-associated weight gain remain poorly characterized. Large cohort studies with >5-year follow-up are needed.
- 4. HIV cure and remission strategies:** bNAb combinations and CAR-T approaches show early promise but remain far from clinical application. Larger, longer trials with diverse populations are required. Standardized definitions of "functional cure" and "remission" are needed for trial design.
- 5. Implementation science for differentiated service delivery at scale:** While DSD models show efficacy in trials, evidence on sustainable implementation, cost-effectiveness across diverse health systems, and integration with other disease programmes (TB, hepatitis, NCDs) is incomplete.
- 6. Youth-specific evidence:** The 2026 WHO case study on youth engagement highlights the paucity of high-quality evidence on youth-specific interventions, preferences, and outcomes. Adolescent-focused RCTs and implementation studies are a priority.
- 7. HIV and ageing:** As ART extends life expectancy, evidence on managing HIV in older adults — including polypharmacy, comorbidity burden, and age-appropriate service delivery — is insufficient.
- 8. Point-of-care viral load and resistance testing:** Rapid, affordable point-of-care platforms for viral load and genotypic resistance testing could transform monitoring in resource-limited settings but remain in early development or limited rollout.

9. **Hepatitis B co-infection management during ART switches:** Switching from TDF-containing to non-TDF regimens risks hepatitis B flares. Prospective data on safe switching strategies are limited.
10. **Mpox and other emerging co-infections in people living with HIV:** The 2022–2024 mpox outbreaks highlighted the vulnerability of immunosuppressed people living with HIV. Evidence on optimal prevention and treatment of mpox and other emerging infections in this population is evolving.

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact	Updated Recommendation
Oral PrEP (TDF-based) as the primary pharmacological prevention option	PURPOSE 1 & 2 trials demonstrate 96% efficacy of twice-yearly subcutaneous lenacapavir; HPTN 084-OLE confirms real-world effectiveness of CAB-LA	Expand PrEP choice menu: Twice-yearly lenacapavir and bimonthly CAB-LA injections should be offered alongside oral PrEP as equivalent first-line options, with choice guided by patient preference, access, and clinical suitability. (<i>GRADE: High; Strength: Strong</i>)
DTG-based regimens as preferred first-line ART for adults and children	DOLPHIN-2 confirms efficacy and safety of DTG dispersible tablets in children < 20 kg; updated periconception safety meta-analysis confirms very low absolute NTD risk	Reinforce DTG as universal preferred first-line across all ages including neonates ≥ 4 weeks and women of childbearing potential, with informed counselling on the very low absolute NTD risk. (<i>GRADE: High; Strength: Strong</i>)
Long-acting injectable CAB + RPV every 2 months as maintenance ART option	ATLAS-2M 256-week data confirm durable suppression and safety	Strengthen recommendation for long-acting injectable ART as a preferred maintenance option for virally suppressed adults who prefer non-daily regimens. (<i>GRADE: High; Strength: Strong</i>)
Single-dose liposomal amphotericin B as preferred induction for cryptococcal meningitis	AMBITION-cm 2-year follow-up confirms sustained mortality benefit	No change to recommendation; evidence strengthens existing guidance. Consider upgrading implementation priority in national programmes. (<i>GRADE: High; Strength: Strong</i>)
Reflex CrAg screening recommended for patients with CD4 < 100	Stepped-wedge trial shows 40% reduction in CM incidence and 28% mortality reduction when reflex CrAg screening applied at CD4 < 200	Expand CrAg reflex screening threshold from CD4 < 100 to CD4 < 200 cells/mm³ in all ART programmes. (<i>GRADE: Moderate; Strength: Strong</i>)

Differentiated service delivery models for stable ART patients	SEARCH-Youth trial shows significant improvement in viral suppression and retention among youth with youth-friendly DSD	Recommend youth-specific DSD models (peer navigators, flexible hours, digital tools) as a standard component of HIV service delivery for adolescents and young adults aged 15–24. (<i>GRADE: Moderate; Strength: Conditional</i>)
Digital health interventions (SMS, apps) to support adherence	Meta-analysis of 18 RCTs confirms 8-percentage-point improvement in viral suppression	Upgrade from conditional to strong recommendation for integration of mHealth adherence interventions, particularly in sub-Saharan Africa and among youth populations. (<i>GRADE: Moderate; Strength: Strong</i>)
No prior recommendation on once-weekly oral ART	Phase II data on islatravir/lenacapavir once-weekly show 90% suppression in treatment-experienced patients	New conditional recommendation: Once-weekly oral ART combinations may be considered for treatment-experienced patients with adherence challenges, pending Phase III confirmation. (<i>GRADE: Moderate; Strength: Conditional</i>)
No prior recommendation on bNAb-based therapeutic strategies	Phase II VRC01LS + 10-1074 trial shows 62% maintained suppression during ART interruption	No recommendation for clinical use at this time. Evidence is promising but insufficient for a population-level recommendation. Encourage enrolment in clinical trials and continued research investment. (<i>GRADE: Moderate for proof-of-concept; insufficient for clinical recommendation</i>)
Xpert MTB/RIF Ultra as preferred TB diagnostic in advanced HIV	TB-LAMP diagnostic accuracy study confirms inferior sensitivity vs Xpert Ultra	No change to recommendation. Xpert Ultra remains the preferred rapid TB diagnostic. TB-LAMP may be considered only as a triage tool where Xpert is unavailable. (<i>GRADE: Moderate; Strength: Conditional for TB-LAMP as triage only</i>)
Youth engagement in guideline processes (new 2026 recommendation)	2026 WHO case study provides framework but limited outcome data	Adopt youth engagement frameworks in all future WHO HIV guideline development cycles, with built-in evaluation of impact on recommendation relevance and programme uptake. (<i>GRADE: Very Low; Strength: Conditional</i>)

This dynamic guideline report synthesizes the five identified WHO HIV guidelines and post-guideline evidence through 29 April 2026. All GRADE ratings follow the standard framework for quality of evidence and strength of recommendation. Clinical decision-making should integrate these recommendations with local epidemiology, health system capacity, patient values, and resource availability. This report should be updated as new evidence emerges, particularly from ongoing Phase III trials of lenacapavir PrEP, once-weekly oral ART, and HIV cure strategies.